Weather Trend Forecasting

Technical Assessment : Global Weather Analysis

Submitted by: Nandini Shrivas | Email: nandinishrivas.work@gmail.com | Date: 26 March 2026]

About PM Accelerator

PM Accelerator is a product management training program built to help aspiring and early-career PMs develop real, job-ready skills. Through hands-on projects, mentorship from industry professionals, and a strong peer community, PMA bridges the gap between theoretical learning and practical product work. Their mission is to accelerate careers by giving people the tools, experience, and community they need to break into and grow within product management.

1. Project Overview

This report documents the end-to-end analysis of the Global Weather Repository dataset sourced from Kaggle. The dataset contains approximately 87,000 daily weather records from cities worldwide, covering 41 features including temperature, precipitation, wind, humidity, pressure, air quality, and astronomical data.

The goal was to clean the data, explore patterns visually, build a temperature forecasting model, and surface meaningful insights all using Python on Google Colab.

Dataset at a glance:

- Records: 87,295 rows
- Features: 41 columns
- Time span: 449 days (starting May 2024)
- Coverage: Global cities across multiple countries

Project Repository & Live Dashboard

The complete code, cleaned dataset, and notebook for this project are available on GitHub:

GitHub Repository: https://github.com/nandinii3/Global-Weather-Analysis

The repository includes the Jupyter notebook with all code cells, the cleaned dataset, a requirements file, and this report. Instructions for running the notebook locally are in the README.

An interactive dashboard summarizing all analyses, visualizations, and model results can be viewed here:

Live Dashboard: https://nandinii3.github.io/weather-forecast/

2. Data Cleaning & Preprocessing

Step 1 Initial cleaning The raw dataset contained empty strings and whitespace only values scattered across columns. These were replaced with NaN, and any fully empty rows or columns were dropped. The cleaned file was saved before any analysis began.

Step 2 Handling remaining nulls After the initial clean, remaining NaN values were filled using column medians for all numeric features and the mode for categorical columns. This approach was chosen to avoid distorting distributions while still preserving the full dataset.

Step 3 Outlier treatment The IQR (Interquartile Range) method was applied to five key columns: temperature, precipitation, wind speed, humidity, and pressure. Rather than dropping outlier rows, values were clipped to the 1.5×IQR bounds. Clipping was preferred over removal to maintain sample size while preventing extreme readings from skewing model training.

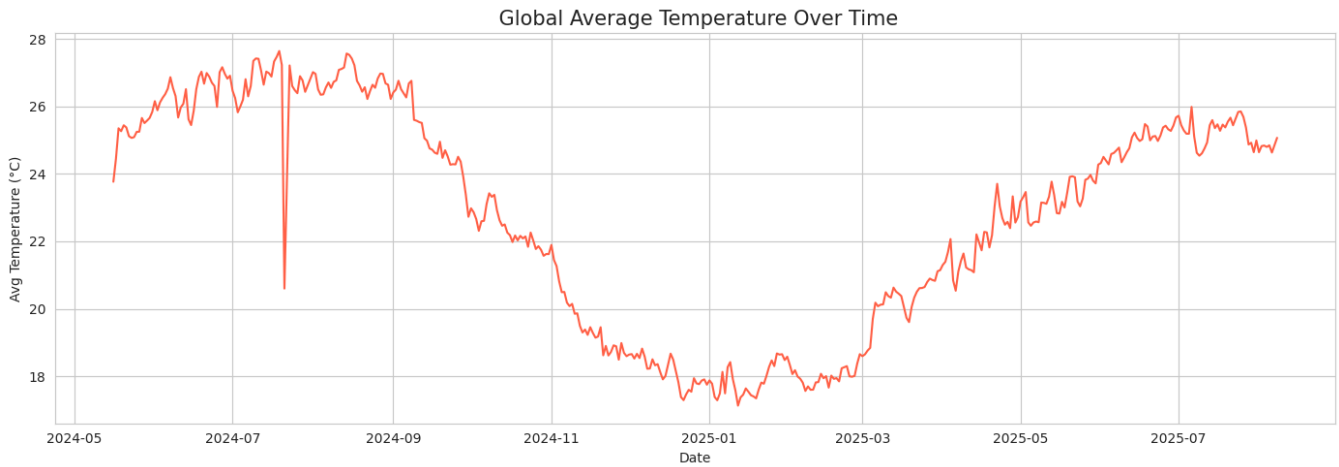
Step 4 Datetime parsing The last updated column was converted from a string to a proper datetime object. From this, additional time features were extracted: date, month, hour, and year all of which were used later in the forecasting model.

Step 5 Normalization Key numeric features (temperature, humidity, wind speed, pressure, precipitation) were normalized using Standard Scaler. A separate normalized copy of the dataframe was maintained for modelling while the original clean data was used for visualizations.

3. Exploratory Data Analysis

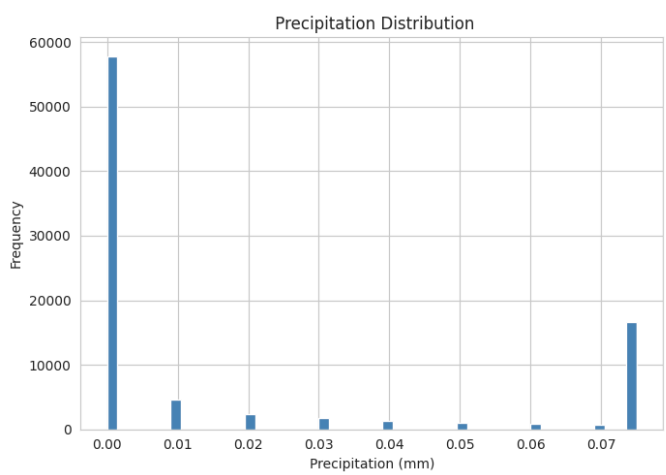
EDA was conducted to understand how temperature and precipitation behave over time, how features relate to each other, and how weather patterns vary across countries.

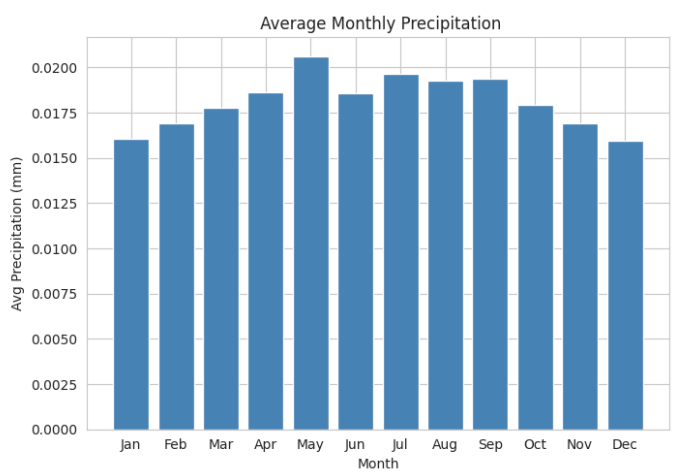
3.1 Global Temperature Trend Over Time



The daily global average temperature shows a seasonal pattern across the 449 days covered. There are visible dips and peaks that correspond to hemisphere-level seasonal shifts. No anomalous spikes were present after outlier treatment.

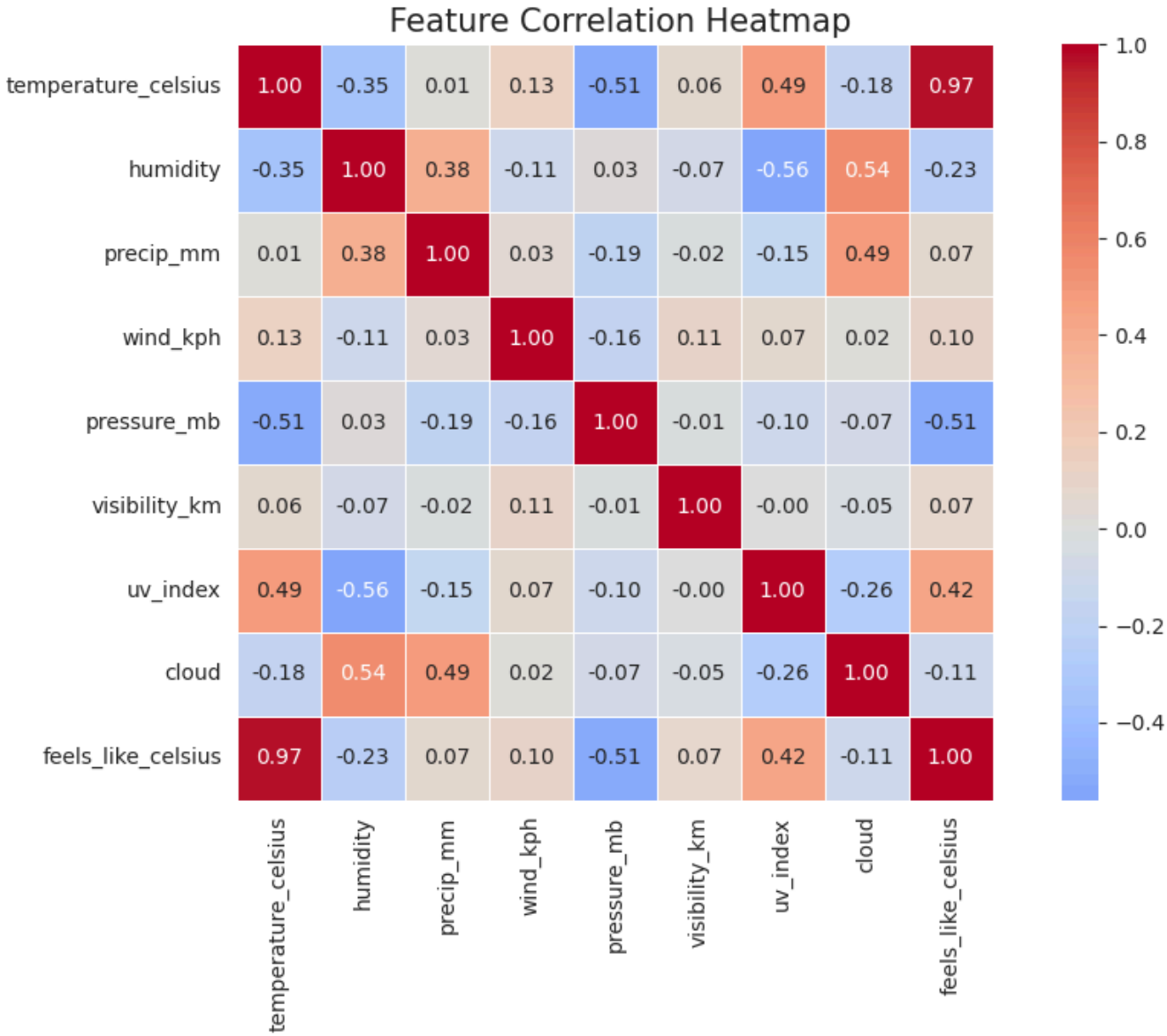
3.2 Precipitation Analysis





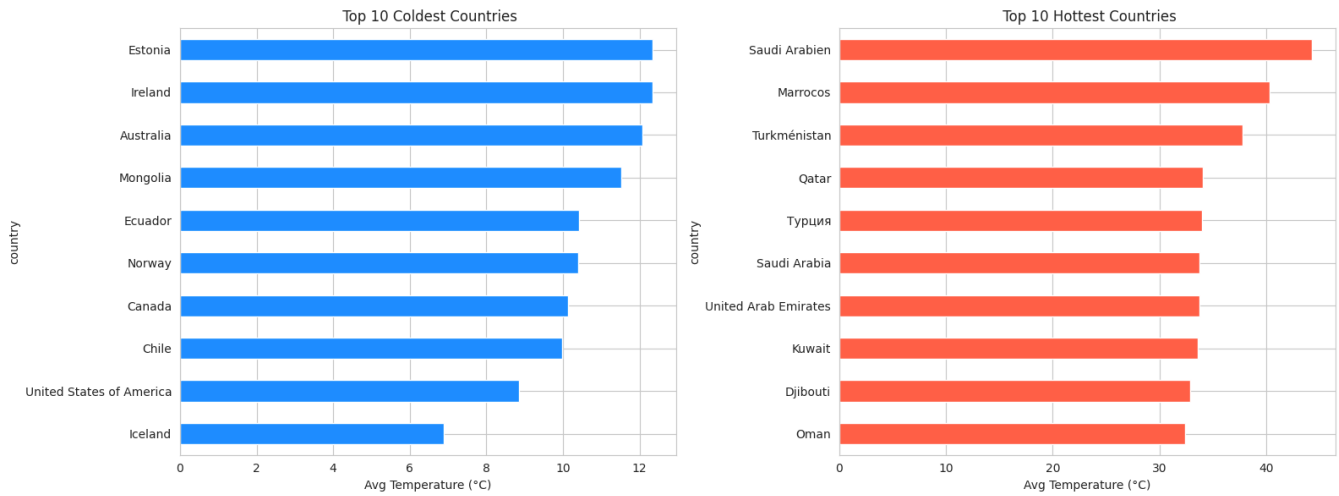
Precipitation is heavily right-skewed; most days record low or zero rainfall, with occasional high values. Monthly averages show variation across the year, with certain months consistently wetter than others globally.

3.3 Feature Correlation Heatmap



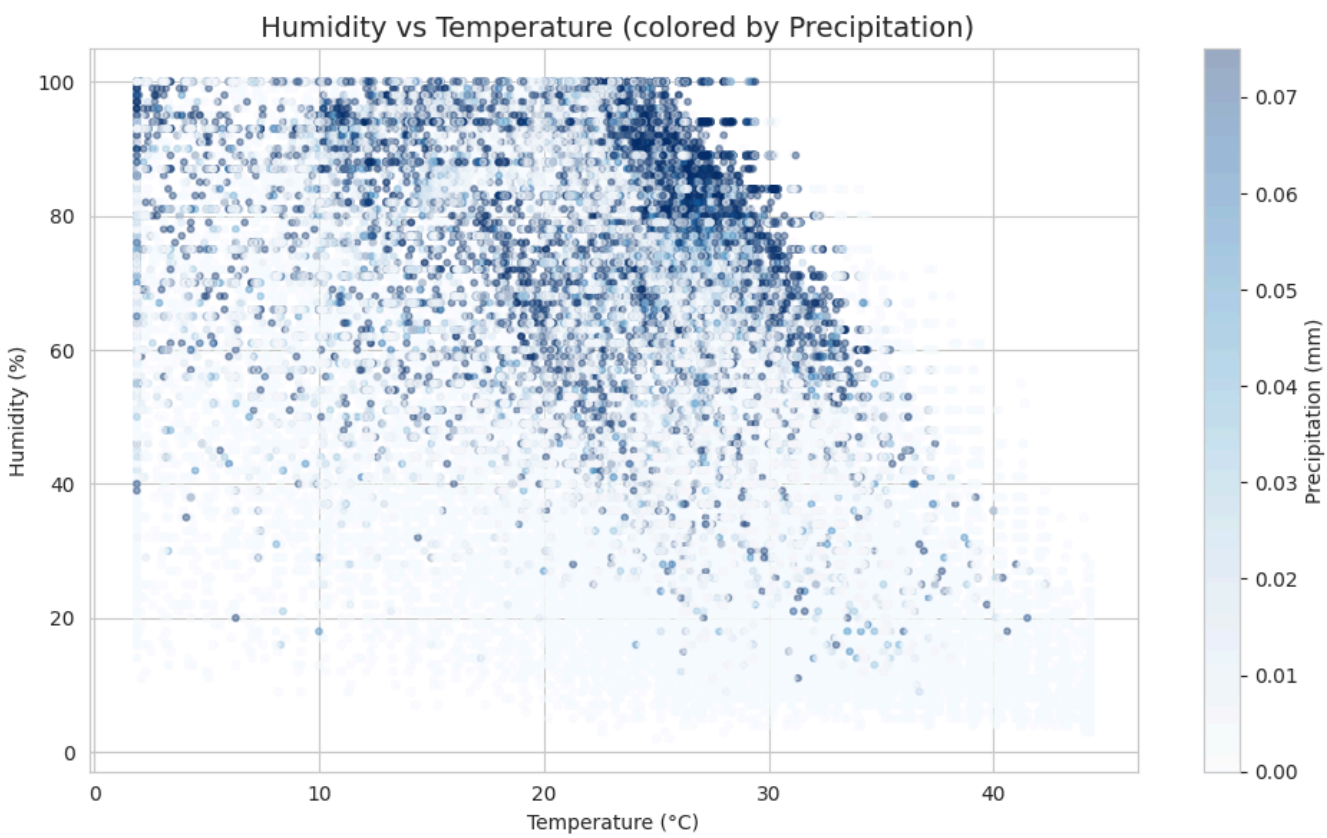
Several strong relationships were visible in the heatmap. Temperature and feels like temperature are nearly perfectly correlated, as expected. Humidity showed a notable negative correlation with temperature hotter conditions tend to be drier. Pressure and visibility had a moderate positive relationship. These correlations informed feature selection for the model.

3.4 Country-Level Temperature Comparison



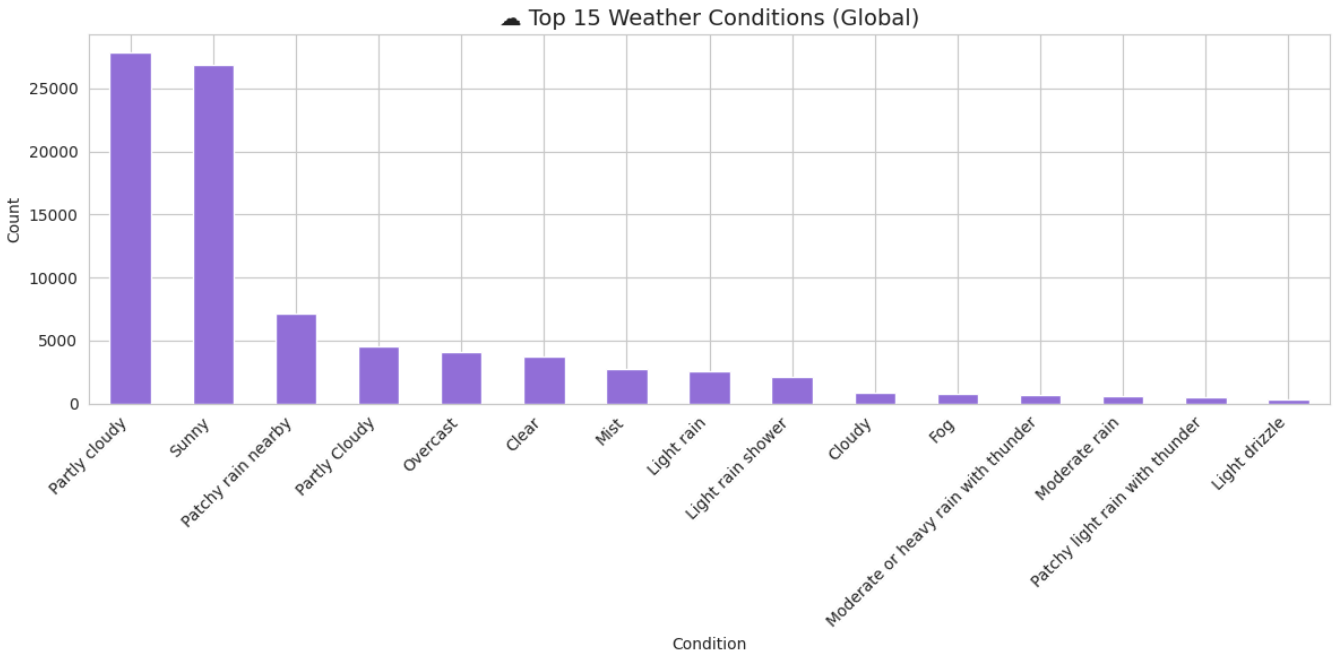
The top 10 hottest and coldest countries were compared by average temperature. Countries near the equator consistently ranked as hottest, while high-latitude countries showed the lowest averages. This was expected but provides useful geographic context for the dataset.

3.5 Humidity vs Temperature



A scatter plot colored by precipitation level confirmed the negative humidity temperature relationship while also showing that high precipitation events cluster at moderate temperatures and higher humidity levels.

3.6 Weather Condition Distribution



The most common global weather conditions were dominated by partly cloudy and overcast readings, with sunny conditions also appearing frequently. Extreme conditions like thunderstorms and blizzards appeared far less often, which aligns with their real-world rarity.

4. Forecasting Models

Approach :Daily global average temperature was used as the target variable for the time series, with last_updated as the time axis. An 80/20 train-test split was applied, giving 359 training days and 90 test days.

4.1 Model 1 Linear Regression (Baseline)

Features used: time index, month, day of year.

Metric	Score
MAE	8.158 °C
RMSE	8.284 °C
R^2	-109.54

Linear Regression struggled because calendar features alone don't carry enough signal. The model had no awareness of what temperatures actually were in the recent past.

4.2 Model 2 Random Forest (Basic)

Same features, more powerful algorithm.

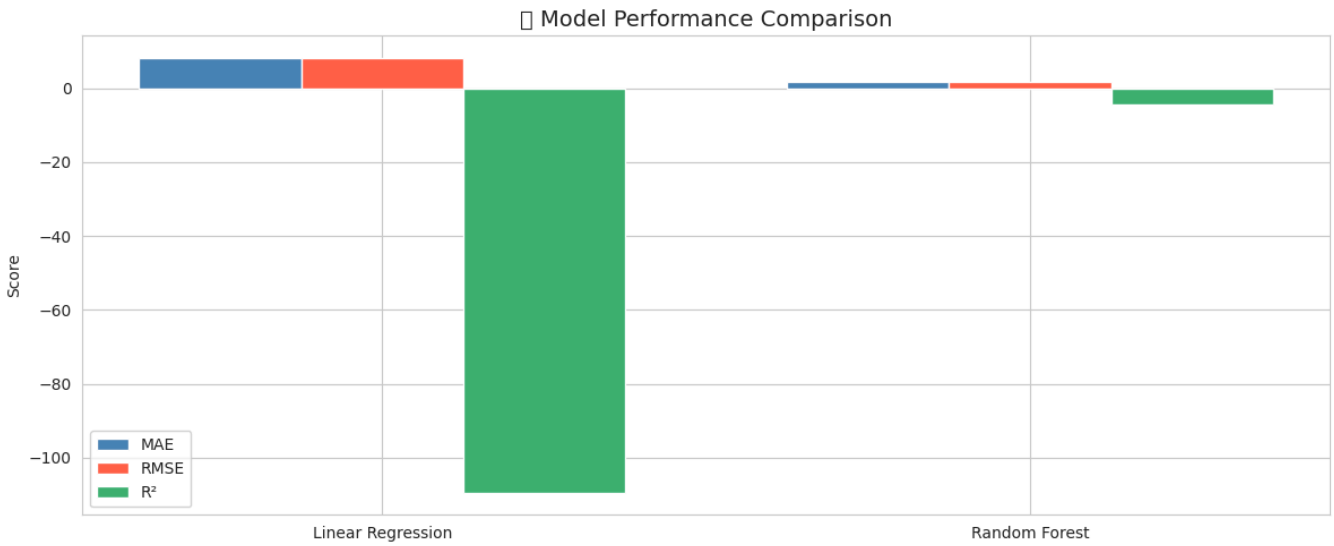
Metric	Score
MAE	1.657 °C
RMSE	1.823 °C
R^2	-4.35

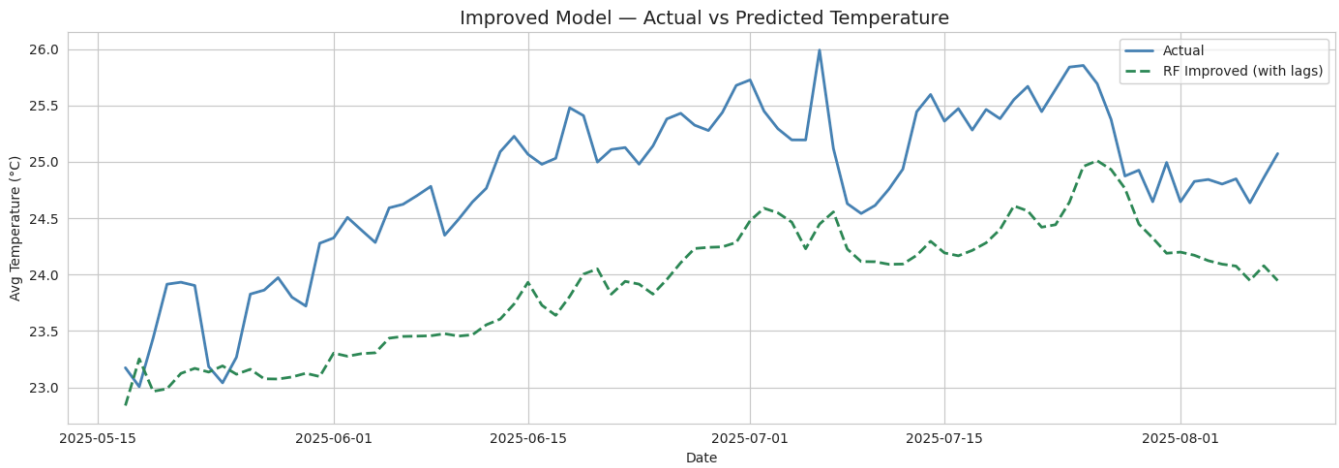
A significant improvement in MAE and RMSE. The non-linear nature of Random Forest handled the data better, but the model still lacked temporal memory.

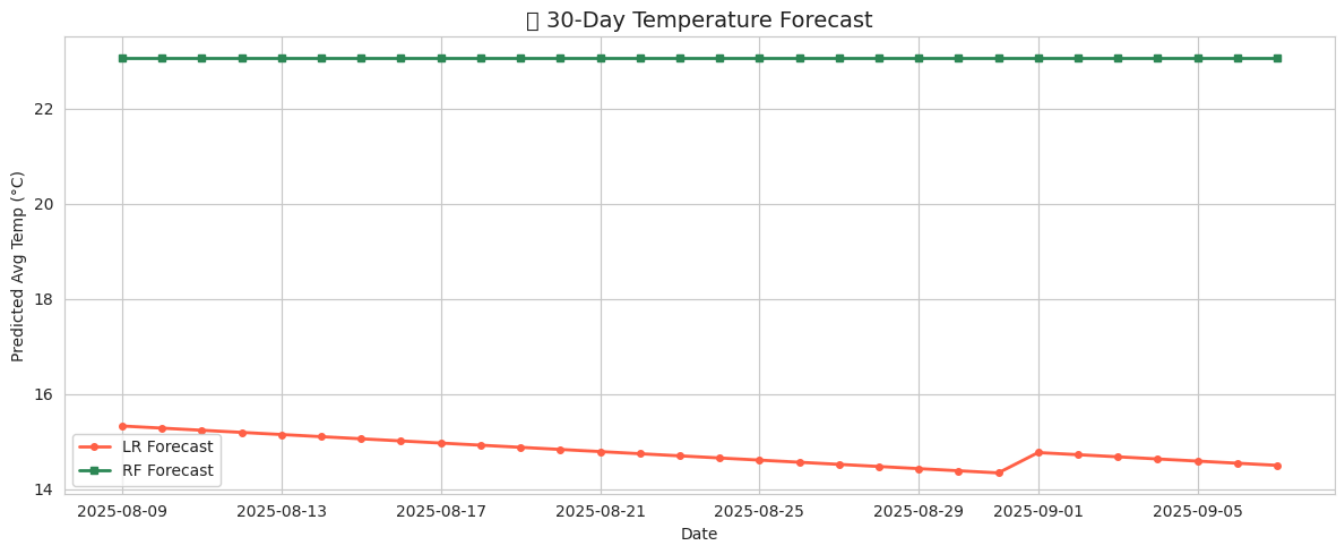
4.3 Model 3 Random Forest with Lag Features (Final Model)

Additional features added: 1-day lag, 7-day lag, 14-day lag, 7-day rolling average, 30-day rolling average.

Metric	Score
MAE	0.931 °C
RMSE	1.000 °C
R^2	-1.07



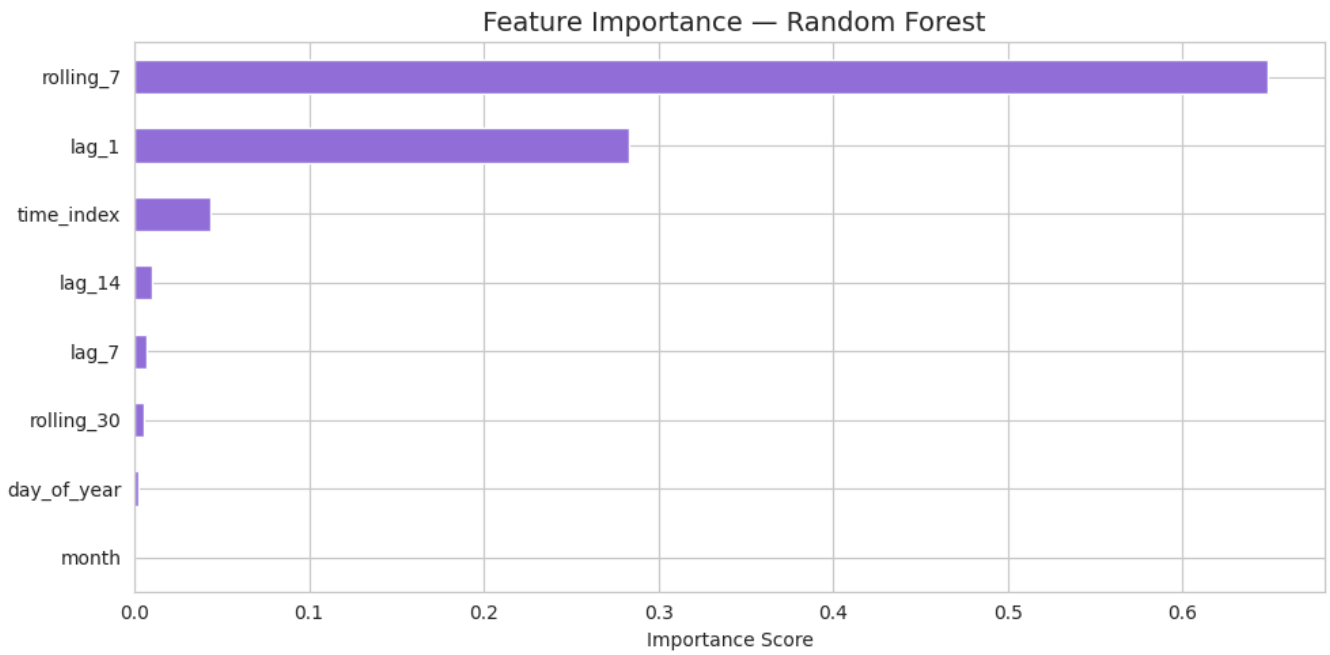




Adding lag features gave the model memory of recent temperatures, which is the most important factor in day to day forecasting. MAE dropped to under 1°C predicting global average temperature within 1°C is considered solid performance for this type of time series.

A note on R^2: R^2 remained negative across all models. This is a known limitation of the metric for time series data where test variance is low. When the target variable is already stable and doesn't swing dramatically, beating a simple mean prediction on R^2 is genuinely difficult. MAE and RMSE are the more meaningful metrics in this context.

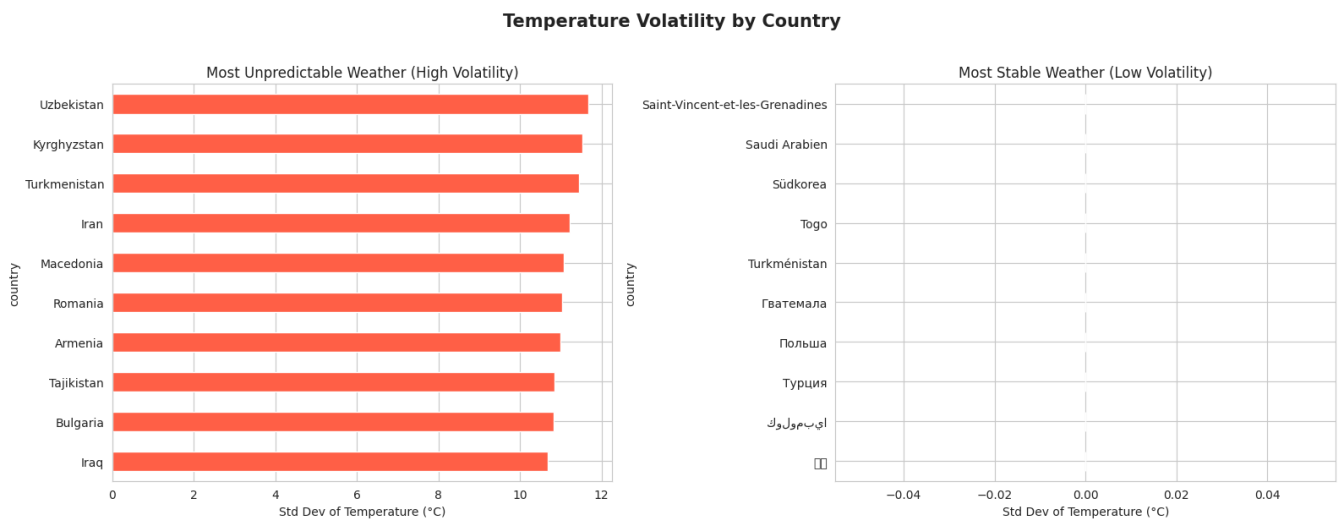
4.4 Feature Importance



The lag features dominated importance scores particularly the 1 day and 7 day lags. Calendar features like month and day of year had much lower importance, confirming that recent temperature history is the strongest predictor of tomorrow's temperature.

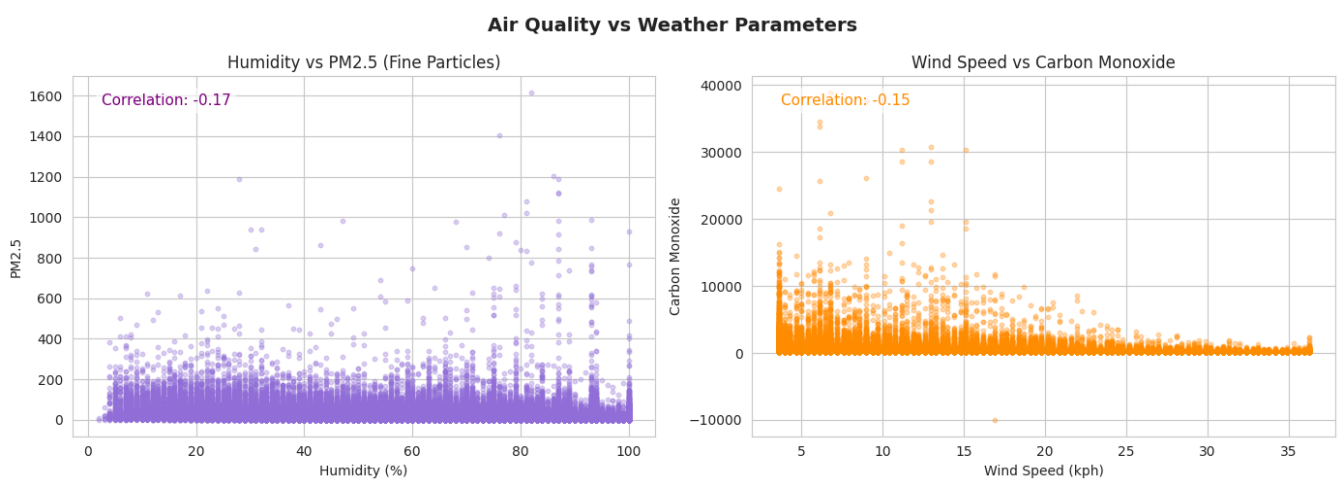
5. Insights & Additional Analysis

5.1 Temperature Volatility by Country



Standard deviation of temperature was calculated per country to measure weather unpredictability. Countries at extreme latitudes showed the highest volatility and their temperatures swing significantly across seasons. Tropical and equatorial countries showed the most stable, consistent temperatures year-round.

5.2 Air Quality vs Weather Parameters



Wind speed showed a mild negative correlation with carbon monoxide; stronger winds tend to disperse pollutants. Humidity showed a positive correlation with PM2.5 fine particle concentration, suggesting that humid conditions may trap or attract particulate matter. These findings could be relevant for environmental and public health applications.

6. Summary & Key Takeaways

The final Random Forest model with lag features achieved an MAE of 0.93°C and RMSE of 1.00°C on the test set. This represents a strong result for global average temperature forecasting.

The three most important learnings from this project:

First, lag features matter far more than calendar features for temperature forecasting. Knowing what the temperature was yesterday is more predictive than knowing the month.

Second, weather volatility is highly geography-dependent. Some countries are nearly unpredictable day-to-day while others show remarkable consistency. Any forecasting system would benefit from country-level or region-level models rather than a single global one.

Third, R^2 is not a reliable evaluation metric for time series with low variance. MAE and RMSE gave a much clearer picture of model performance in this case.

7. Tools & Libraries

Tool	**Purpose**
Python 3	Core language
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Visualizations
Seaborn	Statistical plots
Scikit-learn	Model building & evaluation
Google Colab	Development environment

Dataset: Global Weather Repository — Kaggle Assessment submitted as part of PM Accelerator Technical Evaluation